

# BUSINESS EXCELLENCE ANALYTICS REPORT

## (Furniture, Office Supplies & Technology Performance Analysis)

### 1. Project Overview and Objective

The dataset contains common data quality issues such as missing values, duplicate records, incorrect data entries, and inconsistent formatting. Using Excel, Power Query, and Power BI, the data is cleaned, transformed, and analyzed. Since the dataset consists of multiple related tables, data modeling is performed by creating relationships between the tables. DAX measures, KPI Cards, Gauge Charts, Matrix, and other visualizations are used to build an interactive dashboard for sales analysis and business decision-making.

### 2. Data Sources

- **Source Description and Timeline:**Google Dataset Search and (2023 - 2025)
- **Domain:** Retail

### 3. Problem Statement

This project analyzing Retail Sales data from 2023 to 2025 to identify sales trends, customer purchasing patterns, product performance, and profit insights

### 4. Attribute (Column /Features) Details:

#### Sales Sheet 1:

| Attribute Name | Data Type        | Description                              |
|----------------|------------------|------------------------------------------|
| Order ID       | Integer / String | Unique identifier for each orders        |
| Order Date     | Date Data Type   | Order Date for each orders               |
| Ship Date      | Date Data Type   | Shipped date for each orders             |
| Customer ID    | Integer / String | Unique identifier for each Customers     |
| Product ID     | Integer / String | Unique identifier for each product       |
| Sales          | Decimal Number   | Sales amount for each sales              |
| Quantity       | Whole Number     | Quantity for each Orders                 |
| Discount       | Decimal Number   | Discount for each Orders                 |
| Ship Mode      | Text/Categorical | Shipping mode used for deliver the order |
| Sales Channel  | Text             | Sales channel used for order             |
| Returned       | Categorical      | Shows whether the sale was returned      |

### Customer Sheet 2:

| Attribute Name  | Data Type        | Description                          |
|-----------------|------------------|--------------------------------------|
| Customer ID     | Integer / String | Unique identifier for each Customers |
| Segment         | Categorical      | Segment for each customer type       |
| State           | Categorical      | State of the each customer           |
| Customer Rating | Categorical      | Rating given by the customer         |

### Product Sheet 3:

| Attribute Name | Data Type        | Description                        |
|----------------|------------------|------------------------------------|
| Product ID     | Integer / String | Unique identifier for each product |
| Category       | Categorical      | Category for each product          |
| Sub-Category   | Categorical      | Sub Category for each product      |
| Product Name   | Text             | Name of each Product               |

## 5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

## 6. Data Pre-Processing (Excel / Power Query)

### Tasks Performed:

**1.Data Cleaning & Transformation:** Raw data changed to standardized and structured table format

|    | A        | B          | C          | D           | E          | F       | G        | H        | I              | J             | K        |
|----|----------|------------|------------|-------------|------------|---------|----------|----------|----------------|---------------|----------|
|    | Order ID | Order Date | Ship Date  | Customer ID | Product ID | Sales   | Quantity | Discount | Ship Mode      | Sales Channel | Returned |
| 1  | ORD00001 | 01/26/2023 | 2023-01-29 | CUST0211    | PROD002    | ₹ 4,075 | 6        | 0.2      |                | mobile App    | yes      |
| 2  | ORD00002 | 10/25/2023 | 2023-10-29 | CUST0178    | PROD049    | ₹ 4,035 | 5        | 0        | First Class    | mobile App    | no       |
| 3  | ORD00003 | 25-11-2024 | 2024-11-30 | CUST0069    | PROD053    | ₹ 1,676 | 10       | 0.1      |                | Mobile App    | yes      |
| 4  | ORD00004 | 12/01/2024 | 2024-12-04 | CUST0053    | PROD080    | ₹ 4,538 | 3        | 0.3      | First Class    | Online        | yes      |
| 5  | ORD00005 | 11/16/2023 | 2023-11-18 | CUST0157    | PROD004    | ₹ 783   | 7        | 0.1      | second Class   | store         | no       |
| 6  | ORD00006 | 25-11-2024 | 2024-11-30 | CUST0118    | PROD012    | ₹ 1,026 | 9        | 0.1      | Standard Class | Store         | no       |
| 7  | ORD00007 | 01/07/2025 | 2025-01-10 | CUST0202    | PROD007    | ₹ 352   | 2        | 0        | second Class   | Store         | yes      |
| 8  | ORD00008 | 09/20/2024 | 2024-09-21 | CUST0186    | PROD036    | ₹ 2,064 | 3        | 0.3      | second Class   | Store         | no       |
| 9  | ORD00009 | 02-08-2023 | 2023-08-09 | CUST0020    | PROD022    | ₹ 3,637 | 5        | 0.2      | Standard Class | Store         | no       |
| 10 | ORD00010 | 07/16/2023 | 2023-07-17 | CUST0177    | PROD028    | ₹ 569   | 8        | 0.1      | second Class   | online        | no       |
| 11 | ORD00011 | 01/11/2023 | 2023-01-16 | CUST0004    | PROD068    | ₹ 4,248 | 4        | 0.3      |                | Mobile App    | no       |
| 12 | ORD00012 | 26-03-2024 | 2024-03-30 | CUST0130    | PROD077    | ₹ 1,533 | 1        | 0.1      | First Class    | Store         | no       |
| 13 | ORD00013 | 03/06/2025 | 2025-03-12 | CUST0027    | PROD085    | ₹ 1,616 | 5        | 0.1      | second Class   | Store         | yes      |
| 14 | ORD00014 | 02/27/2024 | 2024-02-29 | CUST0028    | PROD035    | ₹ 2,865 | 3        | 0.3      | First Class    | Online        | yes      |
| 15 | ORD00015 | 05-05-2024 | 2024-05-08 | CUST0223    | PROD038    | ₹ 3,781 | 10       | 0.1      | second Class   | online        | no       |
| 16 | ORD00016 | 09/15/2023 | 2023-09-21 | CUST0223    | PROD009    | ₹ 827   | 6        | 0.2      | second Class   | Mobile App    | no       |
| 17 | ORD00017 | 10/05/2023 | 2023-10-06 | CUST0239    | PROD066    | ₹ 4,480 | 10       | 0        | second Class   | online        | no       |
| 18 | ORD00018 | 15-04-2025 | 2025-04-20 | CUST0006    | PROD042    | ₹ 543   | 3        | 0.1      | First Class    | Mobile App    | no       |
| 19 | ORD00019 | 04/23/2023 | 2023-04-30 | CUST0135    | PROD075    | ₹ 2,608 | 2        | 0.1      | second Class   | Mobile App    | yes      |

|    | A           | B           | C           | D               |
|----|-------------|-------------|-------------|-----------------|
|    | Customer ID | Segment     | State       | Customer Rating |
| 1  | CUST0001    | Home Office | MAHARASHTRA | 4               |
| 2  | CUST0002    | consumer    | telangana   | 1               |
| 3  | CUST0003    | Consumer    | tElanganA   | 5               |
| 4  | CUST0004    | Consumer    | Kerala      | 3               |
| 5  | CUST0005    | home Office | maharashtra | 4               |
| 6  | CUST0006    | Corporate   | Tamil Nadu  | 4               |
| 7  | CUST0007    | Consumer    | TAMIL NADU  | 3               |
| 8  | CUST0008    | Home Office | karnataka   | 2               |
| 9  | CUST0009    | Home Office | Tamil Nadu  | 5               |
| 10 | CUST0010    | Corporate   | Tamil Nadu  | 5               |
| 11 | CUST0011    | consumer    | Karnataka   | 4               |
| 12 | CUST0012    | Home Office | Kerala      | 1               |
| 13 | CUST0013    | Home Office | Maharashtra | 2               |
| 14 | CUST0014    | corporate   | Karnataka   | 1               |
| 15 | CUST0015    | Corporate   | Maharashtra | 2               |
| 16 | CUST0016    | Consumer    | Karnataka   | 3               |
| 17 | CUST0017    | corporate   | Telangana   | 5               |
| 18 | CUST0018    | Consumer    | Maharashtra | 3               |
| 19 | CUST0019    | Home Office | Tamil Nadu  | 3               |

| Product ID | Category        | Sub-Category | Product Name |
|------------|-----------------|--------------|--------------|
| PROD001    | Office Supplies | Accessories  | Product 1    |
| PROD002    | Office Supplies | Binders      | Product 2    |
| PROD003    | Furniture       | Chairs       | Product 3    |
| PROD004    | Technology      | Binders      | Product 4    |
| PROD005    | Technology      | Storage      | Product 5    |
| PROD006    | Furniture       | Binders      | Product 6    |
| PROD007    | Furniture       | Storage      | Product 7    |
| PROD008    | Furniture       | Phones       | Product 8    |
| PROD009    | Technology      | Storage      | Product 9    |
| PROD010    | Furniture       | Tables       | Product 10   |
| PROD011    | Office Supplies | Chairs       | Product 11   |
| PROD012    | Office Supplies | Chairs       | Product 12   |
| PROD013    | Furniture       | Accessories  | Product 13   |
| PROD014    | Technology      | Chairs       | Product 14   |
| PROD015    | Furniture       | Tables       | Product 15   |
| PROD016    | Office Supplies | Storage      | Product 16   |
| PROD017    | Furniture       | Binders      | Product 17   |
| PROD018    | Office Supplies | Accessories  | Product 18   |
| PROD019    | Office Supplies | Accessories  | Product 19   |

< > Sales Customers **Products** +

## 2.Data Cleaning:

From sales table I have used **LEFT,RIGHT,MID** functions on the **Order ID** Column to Extract the Id text into 3 separate columns

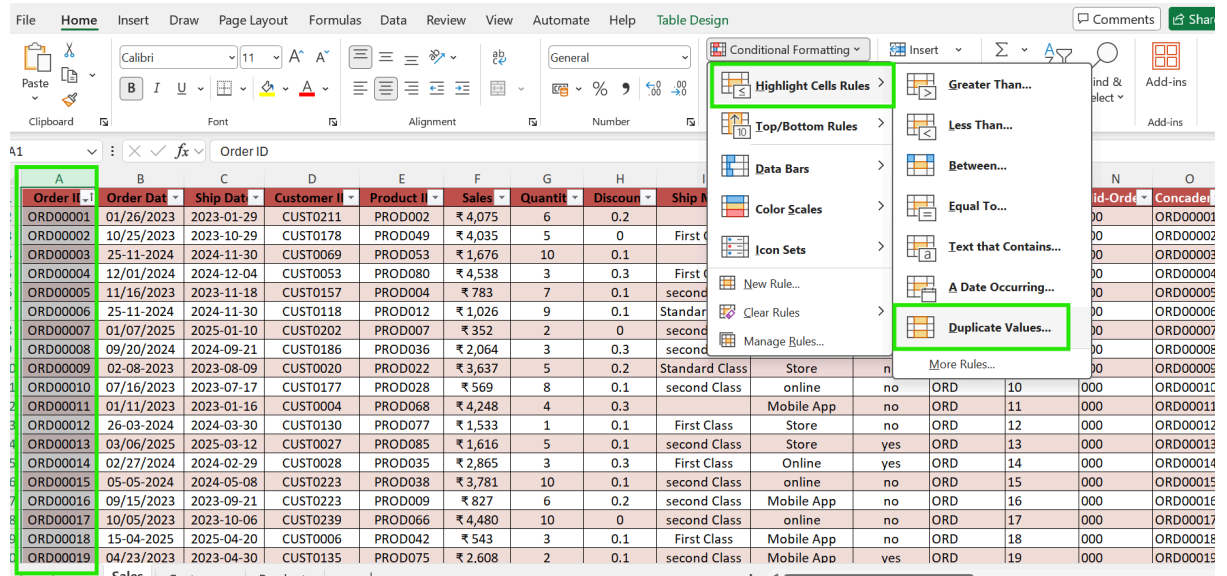
Next I have applied **CONCATENATE** Functions on the those extracted separated columns merged to one single column

Then used the **PROPER** functions on the **RETURNED, SALES CHANNEL,SHIP MODE** Columns to correct the text formatting and capitalize the first letter of each word

| Left - Ord | Right - Ord | Mid-Ord | Concaten | Proper -Returned | Proper - Sales Channel | Proper-Ship mode |
|------------|-------------|---------|----------|------------------|------------------------|------------------|
| ORD        | 01          | 000     | ORD00001 | Yes              | Mobile App             |                  |
| ORD        | 02          | 000     | ORD00002 | No               | Mobile App             | First Class      |
| ORD        | 03          | 000     | ORD00003 | Yes              | Mobile App             |                  |
| ORD        | 04          | 000     | ORD00004 | Yes              | Online                 | First Class      |
| ORD        | 05          | 000     | ORD00005 | No               | Store                  | Second Class     |
| ORD        | 06          | 000     | ORD00006 | No               | Store                  | Standard Class   |
| ORD        | 07          | 000     | ORD00007 | Yes              | Store                  | Second Class     |
| ORD        | 08          | 000     | ORD00008 | No               | Store                  | Second Class     |
| ORD        | 09          | 000     | ORD00009 | No               | Store                  | Standard Class   |
| ORD        | 10          | 000     | ORD00010 | No               | Online                 | Second Class     |
| ORD        | 11          | 000     | ORD00011 | No               | Mobile App             |                  |
| ORD        | 12          | 000     | ORD00012 | No               | Store                  | First Class      |
| ORD        | 13          | 000     | ORD00013 | Yes              | Store                  | Second Class     |
| ORD        | 14          | 000     | ORD00014 | Yes              | Online                 | First Class      |
| ORD        | 15          | 000     | ORD00015 | No               | Online                 | Second Class     |
| ORD        | 16          | 000     | ORD00016 | No               | Mobile App             | Second Class     |
| ORD        | 17          | 000     | ORD00017 | No               | Online                 | Second Class     |
| ORD        | 18          | 000     | ORD00018 | No               | Mobile App             | First Class      |
| ORD        | 19          | 000     | ORD00019 | Yes              | Mobile App             | Second Class     |

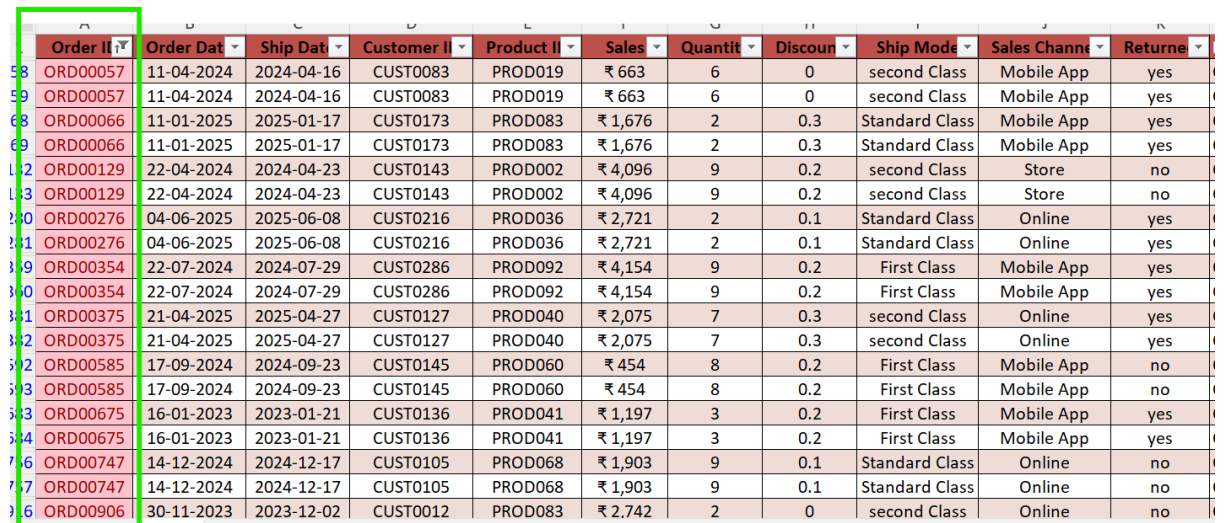
## Find Duplicate values

In the Sales table, I used **Conditional Formatting** and applied the **Duplicate Values** option to the Order ID column to identify duplicate orders

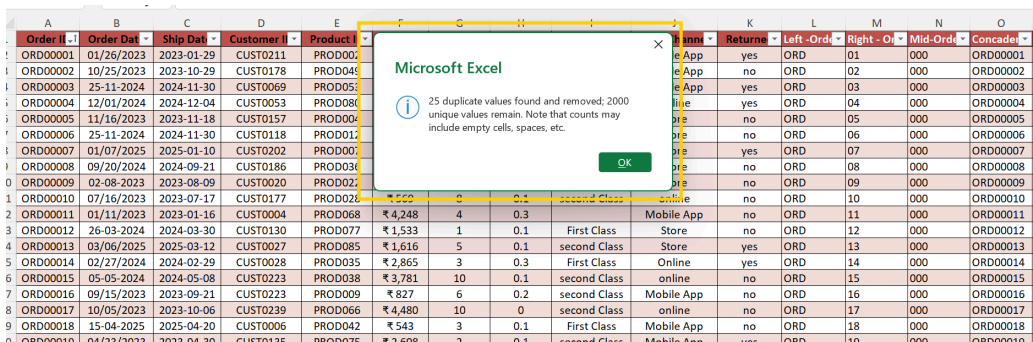
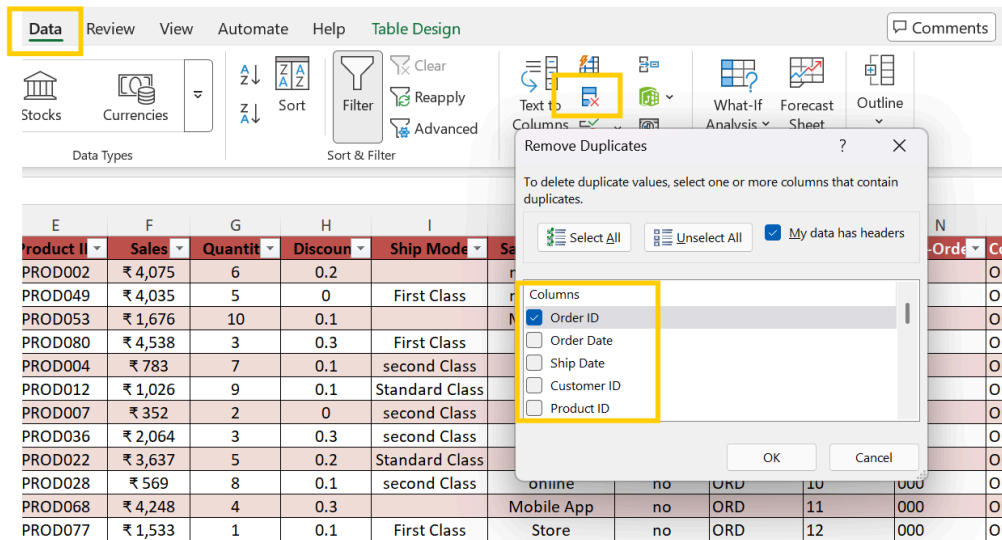


| Order ID | Order Date | Ship Date  | Customer ID | Product ID | Sales   | Quantity | Discount | Ship Mode      | Sales Channel | Return Status |
|----------|------------|------------|-------------|------------|---------|----------|----------|----------------|---------------|---------------|
| ORD00001 | 01/26/2023 | 2023-01-29 | CUST0211    | PROD002    | ₹ 4,075 | 6        | 0.2      | Standard Class | Store         | no            |
| ORD00002 | 10/25/2023 | 2023-10-29 | CUST0178    | PROD049    | ₹ 4,035 | 5        | 0        | First Class    | online        | no            |
| ORD00003 | 25-11-2024 | 2024-11-30 | CUST0069    | PROD053    | ₹ 1,676 | 10       | 0.1      | Standard Class | Mobile App    | yes           |
| ORD00004 | 12/01/2024 | 2024-12-04 | CUST0053    | PROD080    | ₹ 4,538 | 3        | 0.3      | Standard Class | Mobile App    | yes           |
| ORD00005 | 11/16/2023 | 2023-11-18 | CUST0157    | PROD004    | ₹ 783   | 7        | 0.1      | Standard Class | Mobile App    | yes           |
| ORD00006 | 25-11-2024 | 2024-11-30 | CUST0118    | PROD012    | ₹ 1,026 | 9        | 0.1      | Standard Class | Mobile App    | yes           |
| ORD00007 | 01/07/2025 | 2025-01-10 | CUST0202    | PROD007    | ₹ 352   | 2        | 0        | Standard Class | Mobile App    | yes           |
| ORD00008 | 09/20/2024 | 2024-09-21 | CUST0186    | PROD036    | ₹ 2,064 | 3        | 0.3      | Standard Class | Mobile App    | yes           |
| ORD00009 | 02-08-2023 | 2023-08-09 | CUST0020    | PROD022    | ₹ 3,637 | 5        | 0.2      | Standard Class | Store         | no            |
| ORD00010 | 07/16/2023 | 2023-07-17 | CUST0177    | PROD028    | ₹ 569   | 8        | 0.1      | Standard Class | online        | no            |
| ORD00011 | 01/11/2023 | 2023-01-16 | CUST0004    | PROD068    | ₹ 4,248 | 4        | 0.3      | Standard Class | Mobile App    | no            |
| ORD00012 | 26-03-2024 | 2024-03-30 | CUST0130    | PROD077    | ₹ 1,533 | 1        | 0.1      | First Class    | Store         | no            |
| ORD00013 | 03/06/2025 | 2025-03-12 | CUST0027    | PROD085    | ₹ 1,616 | 5        | 0.1      | Standard Class | Store         | yes           |
| ORD00014 | 02/27/2024 | 2024-02-29 | CUST0028    | PROD035    | ₹ 2,865 | 3        | 0.3      | First Class    | Online        | yes           |
| ORD00015 | 05-05-2024 | 2024-05-08 | CUST0223    | PROD038    | ₹ 3,781 | 10       | 0.1      | Standard Class | online        | no            |
| ORD00016 | 09/15/2023 | 2023-09-21 | CUST0223    | PROD009    | ₹ 827   | 6        | 0.2      | Standard Class | Mobile App    | no            |
| ORD00017 | 10/05/2023 | 2023-10-06 | CUST0239    | PROD066    | ₹ 4,480 | 10       | 0        | Standard Class | online        | no            |
| ORD00018 | 15-04-2025 | 2025-04-20 | CUST0006    | PROD042    | ₹ 543   | 3        | 0.1      | First Class    | Mobile App    | no            |
| ORD00019 | 04/23/2023 | 2023-04-30 | CUST0135    | PROD075    | ₹ 2,608 | 2        | 0.1      | Standard Class | Mobile App    | yes           |

Then Filtered the duplicate **Order IDs** using the **Filter by Color** option



| Order ID | Order Date | Ship Date  | Customer ID | Product ID | Sales   | Quantity | Discount | Ship Mode      | Sales Channel | Return Status |
|----------|------------|------------|-------------|------------|---------|----------|----------|----------------|---------------|---------------|
| ORD00001 | 01-04-2024 | 2024-04-16 | CUST0083    | PROD019    | ₹ 663   | 6        | 0        | second Class   | Mobile App    | yes           |
| ORD00007 | 11-04-2024 | 2024-04-16 | CUST0083    | PROD019    | ₹ 663   | 6        | 0        | second Class   | Mobile App    | yes           |
| ORD00066 | 11-01-2025 | 2025-01-17 | CUST0173    | PROD083    | ₹ 1,676 | 2        | 0.3      | Standard Class | Mobile App    | yes           |
| ORD00066 | 11-01-2025 | 2025-01-17 | CUST0173    | PROD083    | ₹ 1,676 | 2        | 0.3      | Standard Class | Mobile App    | yes           |
| ORD00129 | 22-04-2024 | 2024-04-23 | CUST0143    | PROD002    | ₹ 4,096 | 9        | 0.2      | second Class   | Store         | no            |
| ORD00129 | 22-04-2024 | 2024-04-23 | CUST0143    | PROD002    | ₹ 4,096 | 9        | 0.2      | second Class   | Store         | no            |
| ORD00276 | 04-06-2025 | 2025-06-08 | CUST0216    | PROD036    | ₹ 2,721 | 2        | 0.1      | Standard Class | Online        | yes           |
| ORD00276 | 04-06-2025 | 2025-06-08 | CUST0216    | PROD036    | ₹ 2,721 | 2        | 0.1      | Standard Class | Online        | yes           |
| ORD00354 | 22-07-2024 | 2024-07-29 | CUST0286    | PROD092    | ₹ 4,154 | 9        | 0.2      | First Class    | Mobile App    | yes           |
| ORD00354 | 22-07-2024 | 2024-07-29 | CUST0286    | PROD092    | ₹ 4,154 | 9        | 0.2      | First Class    | Mobile App    | yes           |
| ORD00375 | 21-04-2025 | 2025-04-27 | CUST0127    | PROD040    | ₹ 2,075 | 7        | 0.3      | second Class   | Online        | yes           |
| ORD00375 | 21-04-2025 | 2025-04-27 | CUST0127    | PROD040    | ₹ 2,075 | 7        | 0.3      | second Class   | Online        | yes           |
| ORD00585 | 17-09-2024 | 2024-09-23 | CUST0145    | PROD060    | ₹ 454   | 8        | 0.2      | First Class    | Mobile App    | no            |
| ORD00585 | 17-09-2024 | 2024-09-23 | CUST0145    | PROD060    | ₹ 454   | 8        | 0.2      | First Class    | Mobile App    | no            |
| ORD00675 | 16-01-2023 | 2023-01-21 | CUST0136    | PROD041    | ₹ 1,197 | 3        | 0.2      | First Class    | Mobile App    | yes           |
| ORD00675 | 16-01-2023 | 2023-01-21 | CUST0136    | PROD041    | ₹ 1,197 | 3        | 0.2      | First Class    | Mobile App    | yes           |
| ORD00747 | 14-12-2024 | 2024-12-17 | CUST0105    | PROD068    | ₹ 1,903 | 9        | 0.1      | Standard Class | Online        | no            |
| ORD00747 | 14-12-2024 | 2024-12-17 | CUST0105    | PROD068    | ₹ 1,903 | 9        | 0.1      | Standard Class | Online        | no            |
| ORD00906 | 30-11-2023 | 2023-12-02 | CUST0012    | PROD083    | ₹ 2,742 | 2        | 0        | second Class   | Online        | no            |



In the **Data ribbon**, I applied the **Duplicate Values** option to the Order ID column. I identified **25 duplicate values** and removed them.

### 3.Formatting Date Properly

In the Sales table, the **Order Date** and **Ship Date** were not formatted properly. I applied the Delimiter option and used the **DATE, IF**, and **VALUE** functions to convert the dates into the proper **DD-MM-YYYY** format. Now, both the Order Date and Ship Date are cleaned properly

| Order Date formatted | Ship date formatted | Proper Order Date | Proper ship date |
|----------------------|---------------------|-------------------|------------------|
| 01-26-2023           | 2023-01-29          | 26-01-2023        | 29-01-2023       |
| 10-25-2023           | 2023-10-29          | 25-10-2023        | 29-10-2023       |
| 25-11-2024           | 2024-11-30          | 25-11-2024        | 30-11-2024       |
| 12-01-2024           | 2024-12-04          | 01-12-2024        | 04-12-2024       |
| 11-16-2023           | 2023-11-18          | 16-11-2023        | 18-11-2023       |
| 25-11-2024           | 2024-11-30          | 25-11-2024        | 30-11-2024       |
| 01-07-2025           | 2025-01-10          | 07-01-2025        | 10-01-2025       |
| 09-20-2024           | 2024-09-21          | 20-09-2024        | 21-09-2024       |
| 02-08-2023           | 2023-08-09          | 08-02-2023        | 09-08-2023       |
| 07-16-2023           | 2023-07-17          | 16-07-2023        | 17-07-2023       |
| 01-11-2023           | 2023-01-16          | 11-01-2023        | 16-01-2023       |
| 26-03-2024           | 2024-03-30          | 26-03-2024        | 30-03-2024       |
| 03-06-2025           | 2025-03-12          | 06-03-2025        | 12-03-2025       |
| 02-27-2024           | 2024-02-29          | 27-02-2024        | 29-02-2024       |
| 05-05-2024           | 2024-05-08          | 05-05-2024        | 08-05-2024       |
| 09-15-2023           | 2023-09-21          | 15-09-2023        | 21-09-2023       |
| 10-05-2023           | 2023-10-06          | 05-10-2023        | 06-10-2023       |
| 15-04-2025           | 2025-04-20          | 15-04-2025        | 20-04-2025       |

| Proper - Sales Channel | Proper-Ship mode | Order Date formatted | Ship date formatted | Proper Order Date | Proper ship date | Column13 | Column14 | Column15 | Column16 |
|------------------------|------------------|----------------------|---------------------|-------------------|------------------|----------|----------|----------|----------|
| Mobile App             |                  | 01-26-2023           | 2023-01-29          | 26-01-2023        | 29-01-2023       |          |          |          |          |
| Mobile App             | First Class      | 10-25-2023           | 2023-10-29          | 25-10-2023        | 29-10-2023       |          |          |          |          |
| Mobile App             |                  | 25-11-2024           | 2024-11-30          | 25-11-2024        | 30-11-2024       |          |          |          |          |
| Online                 | First Class      | 12-01-2024           | 2024-12-04          | 01-12-2024        | 04-12-2024       |          |          |          |          |
| Store                  | Second Class     | 11-16-2023           | 2023-11-18          | 16-11-2023        | 18-11-2023       |          |          |          |          |
| Store                  | Standard Class   | 25-11-2024           | 2024-11-30          | 25-11-2024        | 30-11-2024       |          |          |          |          |
| Store                  | Second Class     | 01-07-2025           | 2025-01-10          | 07-01-2025        | 10-01-2025       |          |          |          |          |
| Store                  | Second Class     | 09-20-2024           | 2024-09-21          | 20-09-2024        | 21-09-2024       |          |          |          |          |
| Store                  | Standard Class   | 02-08-2023           | 2023-08-09          | 08-02-2023        | 09-08-2023       |          |          |          |          |
| Online                 | Second Class     | 07-16-2023           | 2023-07-17          | 16-07-2023        | 17-07-2023       |          |          |          |          |
| Mobile App             |                  | 01-11-2023           | 2023-01-16          | 11-01-2023        | 16-01-2023       |          |          |          |          |
| Store                  | First Class      | 26-03-2024           | 2024-03-30          | 26-03-2024        | 30-03-2024       |          |          |          |          |
| Store                  | Second Class     | 03-06-2025           | 2025-03-12          | 06-03-2025        | 12-03-2025       |          |          |          |          |
| Online                 | First Class      | 02-27-2024           | 2024-02-29          | 27-02-2024        | 29-02-2024       |          |          |          |          |
| Online                 | Second Class     | 05-05-2024           | 2024-05-08          | 05-05-2024        | 08-05-2024       |          |          |          |          |
| Mobile App             | Second Class     | 09-15-2023           | 2023-09-21          | 15-09-2023        | 21-09-2023       |          |          |          |          |

#### 4.Added Columns

Then, in the Customer table, I applied Nested IF conditions to the Customer Rating column. I categorized the ratings as follows: 5 – Excellent, 4 – Good, 3 – Average, 2 – Poor, and 1 – Very Poor

| Customer ID | Segment     | State       | Customer Rating | Rating status | Proper Segement | Proper State | Column4 | Column5 | Column6 | Column7 | Column8 | Column9 | Column10 |
|-------------|-------------|-------------|-----------------|---------------|-----------------|--------------|---------|---------|---------|---------|---------|---------|----------|
| CUST0001    | Home Office | MAHARASHTRA | 4               | Good          | Home Office     | Maharashtra  |         |         |         |         |         |         |          |
| CUST0002    | consumer    | telangana   | 1               | Very Poor     | Consumer        | Telangana    |         |         |         |         |         |         |          |
| CUST0003    | Consumer    | tElanganA   | 5               | Excellent     | Consumer        | Telangana    |         |         |         |         |         |         |          |
| CUST0004    | Consumer    | Kerala      | 3               | Average       | Consumer        | Kerala       |         |         |         |         |         |         |          |
| CUST0005    | home Office | maharashtra | 4               | Good          | Home Office     | Maharashtra  |         |         |         |         |         |         |          |
| CUST0006    | Corporate   | Tamil Nadu  | 4               | Good          | Corporate       | Tamil Nadu   |         |         |         |         |         |         |          |
| CUST0007    | Consumer    | TAMIL NADU  | 3               | Average       | Consumer        | Tamil Nadu   |         |         |         |         |         |         |          |
| CUST0008    | Home Office | karnataka   | 2               | Poor          | Home Office     | Karnataka    |         |         |         |         |         |         |          |
| CUST0009    | Home Office | Tamil Nadu  | 5               | Excellent     | Home Office     | Tamil Nadu   |         |         |         |         |         |         |          |
| CUST0010    | Corporate   | Tamil Nadu  | 5               | Excellent     | Corporate       | Tamil Nadu   |         |         |         |         |         |         |          |
| CUST0011    | consumer    | Karnataka   | 4               | Good          | Consumer        | Karnataka    |         |         |         |         |         |         |          |
| CUST0012    | Home Office | Kerala      | 1               | Very Poor     | Home Office     | Kerala       |         |         |         |         |         |         |          |
| CUST0013    | Home Office | Maharashtra | 2               | Poor          | Home Office     | Maharashtra  |         |         |         |         |         |         |          |
| CUST0014    | corporate   | Karnataka   | 1               | Very Poor     | Corporate       | Karnataka    |         |         |         |         |         |         |          |
| CUST0015    | Corporate   | Maharashtra | 2               | Poor          | Corporate       | Maharashtra  |         |         |         |         |         |         |          |
| CUST0016    | Consumer    | Karnataka   | 3               | Average       | Consumer        | Karnataka    |         |         |         |         |         |         |          |
| CUST0017    | corporate   | Telangana   | 5               | Excellent     | Corporate       | Telangana    |         |         |         |         |         |         |          |



Then, the text in the **Segment and State columns** was not formatted properly, so I used the **PROPER** function to correct the text formatting.

## 5.Extra Spaces and Inconsistencies

In the Products table, the **Category and Sub-Category columns** text had extra spaces and inconsistencies. So, I used the **TRIM** function to remove the extra spaces and correct the text.

2   

| Product ID | Category        | Sub-Category | Product Name | Trimmed Category | Trimmed Sub-Category | Color |
|------------|-----------------|--------------|--------------|------------------|----------------------|-------|
| PROD001    | Office Supplies | Accessories  | Product 1    | Office Supplies  | Accessories          |       |
| PROD002    | Office Supplies | Binders      | Product 2    | Office Supplies  | Binders              |       |
| PROD003    | Furniture       | Chairs       | Product 3    | Furniture        | Chairs               |       |
| PROD004    | Technology      | Binders      | Product 4    | Technology       | Binders              |       |
| PROD005    | Technology      | Storage      | Product 5    | Technology       | Storage              |       |
| PROD006    | Furniture       | Binders      | Product 6    | Furniture        | Binders              |       |
| PROD007    | Furniture       | Storage      | Product 7    | Furniture        | Storage              |       |
| PROD008    | Furniture       | Phones       | Product 8    | Furniture        | Phones               |       |
| PROD009    | Technology      | Storage      | Product 9    | Technology       | Storage              |       |
| PROD010    | Furniture       | Tables       | Product 10   | Furniture        | Tables               |       |
| PROD011    | Office Supplies | Chairs       | Product 11   | Office Supplies  | Chairs               |       |
| PROD012    | Office Supplies | Chairs       | Product 12   | Office Supplies  | Chairs               |       |
| PROD013    | Furniture       | Accessories  | Product 13   | Furniture        | Accessories          |       |
| PROD014    | Technology      | Chairs       | Product 14   | Technology       | Chairs               |       |
| PROD015    | Furniture       | Tables       | Product 15   | Furniture        | Tables               |       |
| PROD016    | Office Supplies | Storage      | Product 16   | Office Supplies  | Storage              |       |
| PROD017    | Furniture       | Binders      | Product 17   | Furniture        | Binders              |       |
| PROD018    | Office Supplies | Accessories  | Product 18   | Office Supplies  | Accessories          |       |
| PROD019    | Office Supplies | Accessories  | Product 19   | Office Supplies  | Accessories          |       |

Sales Customers **Products**

## 6.Power Query

Created a product-wise median Sales table.Merged it with the Sales table using Product ID.Expanded the median Sales column.Used a Custom Column to fill the missing Sales values with the median.

| Product ID | 1.2 median products |
|------------|---------------------|
| PROD002    | 2064.83             |
| PROD049    | 3321.785            |
| PROD053    | 1676.01             |
| PROD080    | 3422.905            |
| PROD004    | 2911.4              |
| PROD012    | 2031.725            |
| PROD007    | 2514.42             |
| PROD036    | 2676.745            |
| PROD022    | 3070.02             |
| PROD028    | 3016.94             |
| PROD068    | 2761.11             |
| PROD077    | 1884.41             |
| PROD085    | 2211.43             |
| PROD035    | 2120.04             |
| PROD038    | 2760.43             |
| PROD009    | 3199.095            |
| PROD066    | 3189.9              |
| PROD042    | 2383.33             |
| PROD075    | 2713.41             |
| PROD021    | 2379.64             |
| PROD047    | 1421.12             |

**PROPERTIES**

Name  
Table1 (2)

All Properties

**APPLIED STEPS**

Source  
Changed Type  
X median products



Close Query Manage Columns Reduce Rows Sort Transform Combine Parameters Data Sources New Query

Queries [3]  
Table1  
Table1 (2)  
Table1 (3)

fx = Table.ExpandTableColumn(#"Merged Queries", "Table1 (2)", ("Product ID", "median products"),

|    | matted     | Proper Order Date   | Proper ship date | Table1 (2).Product ID | Table1 (2).median products |
|----|------------|---------------------|------------------|-----------------------|----------------------------|
| 1  | 29-01-2023 | 26-01-2023 00:00:00 | 29-01-2023       | PROD002               | 2064.83                    |
| 2  | 11-05-2023 | 04-05-2023 00:00:00 | 11-05-2023       | PROD002               | 2064.83                    |
| 3  | 05-11-2023 | 11-04-2023 00:00:00 | 05-11-2023       | PROD002               | 2064.83                    |
| 4  | 29-10-2023 | 25-10-2023 00:00:00 | 29-10-2023       | PROD049               | 3321.785                   |
| 5  | 11-08-2024 | 04-08-2024 00:00:00 | 11-08-2024       | PROD049               | 3321.785                   |
| 6  | 30-11-2024 | 25-11-2024 00:00:00 | 30-11-2024       | PROD053               | 1676.01                    |
| 7  | 04-12-2024 | 01-12-2024 00:00:00 | 04-12-2024       | PROD080               | 3422.905                   |
| 8  | 01-09-2023 | 28-08-2023 00:00:00 | 01-09-2023       | PROD080               | 3422.905                   |
| 9  | 15-09-2023 | 09-12-2023 00:00:00 | 15-09-2023       | PROD080               | 3422.905                   |
| 10 | 18-11-2023 | 16-11-2023 00:00:00 | 18-11-2023       | PROD004               | 2911.4                     |
| 11 | 30-11-2024 | 25-11-2024 00:00:00 | 30-11-2024       | PROD012               | 2031.725                   |
| 12 | 19-11-2023 | 11-12-2023 00:00:00 | 19-11-2023       | PROD012               | 2031.725                   |
| 13 | 26-02-2025 | 19-02-2025 00:00:00 | 26-02-2025       | PROD012               | 2031.725                   |
| 14 | 03-11-2023 | 27-10-2023 00:00:00 | 03-11-2023       | PROD012               | 2031.725                   |
| 15 | 25-10-2024 | 20-10-2024 00:00:00 | 25-10-2024       | PROD012               | 2031.725                   |
| 16 | 10-01-2025 | 07-01-2025 00:00:00 | 10-01-2025       | PROD007               | 2514.42                    |
| 17 | 21-09-2024 | 20-09-2024 00:00:00 | 21-09-2024       | PROD036               | 2676.745                   |
| 18 | 09-08-2023 | 08-02-2023 00:00:00 | 09-08-2023       | PROD022               | 3070.02                    |
| 19 | 17-07-2023 | 16-07-2023 00:00:00 | 17-07-2023       | PROD028               | 3016.94                    |
| 20 | 16-01-2023 | 11-01-2023 00:00:00 | 16-01-2023       | PROD068               | 2761.11                    |
| 21 |            |                     |                  |                       |                            |

Query Settings

PROPERTIES  
Name  
Table1

APPLIED STEPS  
Source  
Changed Type  
Merged Queries  
Expanded Table1 (2)  
Added Custom  
final sales  
Changed Type1  
Replaced Value  
Capitalized Each Word

File Home Transform Add Column View

Close & Load Refresh Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Whole Number Merge Queries Append Queries Combine Files Manage Parameters Data source settings New Source Recent Sources Enter Data

Queries [3]  
Table1  
Table1 (2)  
Table1 (3)

fx = Table.ReorderColumns(#"Added Custom",("Order ID", "Order Date", "Ship Date", "Customer ID",

|    | Customer ID | Product ID | final sales | Sales   | Quantity |
|----|-------------|------------|-------------|---------|----------|
| 1  | IT0211      | PROD002    | 4075.4      | 4075.4  | 6        |
| 2  | IT0173      | PROD002    | 327.45      | 327.45  | 8        |
| 3  | IT0275      | PROD002    | 1262.75     | 1262.75 | 3        |
| 4  | IT0178      | PROD049    | 4035.22     | 4035.22 | 5        |
| 5  | IT0095      | PROD049    | 3968.01     | 3968.01 | 8        |
| 6  | IT0069      | PROD053    | 1676.01     | 1676.01 | 10       |
| 7  | IT0053      | PROD080    | 4538.4      | 4538.4  | 3        |
| 8  | IT0290      | PROD080    | 1672.66     | 1672.66 | 7        |
| 9  | IT0089      | PROD080    | 3964.58     | 3964.58 | 4        |
| 10 | IT0157      | PROD004    | 783.4       | 783.4   | 7        |
| 11 | IT0118      | PROD012    | 1026.47     | 1026.47 | 9        |
| 12 | IT0046      | PROD012    | 1835.05     | 1835.05 | 9        |
| 13 | IT0008      | PROD012    | 1665.1      | 1665.1  | 5        |
| 14 | IT0060      | PROD012    | 1427.32     | 1427.32 | 6        |
| 15 | IT0020      | PROD012    | 885.43      | 885.43  | 4        |

Query Settings

PROPERTIES  
Name  
Table1

APPLIED STEPS  
Source  
Changed Type  
Merged Queries  
Expanded Table1 (2)  
Added Custom  
final sales  
Changed Type1  
Replaced Value  
Capitalized Each Word

In the Ship Mode column, I identified the null values and replaced them using the mode value, which was First Class

Close Query Manage Columns Reduce Rows Sort Transform Combine Parameters Data Sources New Query

Queries [3]  
Table1  
Table1 (2)  
Table1 (3)

fx = Table.TransformColumns(#"Replaced Value",({"Ship Mode", Text.Proper, type text}))

|    | Sales   | Quantity | Discount | Ship Mode      | Sales Channel |
|----|---------|----------|----------|----------------|---------------|
| 1  | 4075.4  | 6        | 0        | First Class    | mobile App    |
| 2  | 327.45  | 8        | 0        | Second Class   | Online        |
| 3  | 1262.75 | 3        | 0        | First Class    | Online        |
| 4  | 4035.22 | 5        | 0        | First Class    | mobile App    |
| 5  | 3968.01 | 8        | 0        | First Class    | Mobile App    |
| 6  | 1676.01 | 10       | 0        | First Class    | Mobile App    |
| 7  | 4538.4  | 3        | 0        | First Class    | Online        |
| 8  | 1672.66 | 7        | 0        | First Class    | Mobile App    |
| 9  | 3964.58 | 4        | 0        | Standard Class | Mobile App    |
| 10 | 783.4   | 7        | 0        | Second Class   | store         |
| 11 | 1026.47 | 9        | 0        | Standard Class | Store         |
| 12 | 1835.05 | 9        | 0        | First Class    | Mobile App    |
| 13 | 1665.1  | 5        | 0        | Second Class   | Online        |
| 14 | 1427.32 | 6        | 0        | Standard Class | Online        |
| 15 | 885.43  | 4        | 0        | Standard Class | Online        |
| 16 | 352.44  | 2        | 0        | Second Class   | Store         |
| 17 | 2063.72 | 3        | 0        | Second Class   | Store         |
| 18 | 3637.24 | 5        | 0        | Standard Class | Store         |
| 19 | 569     | 8        | 0        | Second Class   | online        |
| 20 | 4248.01 | 4        | 0        | First Class    | Mobile App    |

Query Settings

PROPERTIES  
Name  
Table1

APPLIED STEPS  
Source  
Changed Type  
Merged Queries  
Expanded Table1 (2)  
Added Custom  
final sales  
Changed Type1  
Replaced Value  
Capitalized Each Word

7.I calculated the **Revenue** using the **Sales Amount** and **Discount** columns.

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

J2

Finally I have calculated the number of days between the Order Date and Ship Date to determine the shipping duration

FileHomeInsertDrawTable LayoutFormulasDataViewAutomateHelpTable Design

Paste

Clipboard

Aptos Narrow

11

A

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U

Font

Alignment

General

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Number

Conditional Formatting

Format as Table

Cell Styles

Styles

Insert

Delete

Format

Cells

Editing

Sort & Filter

Find & Select

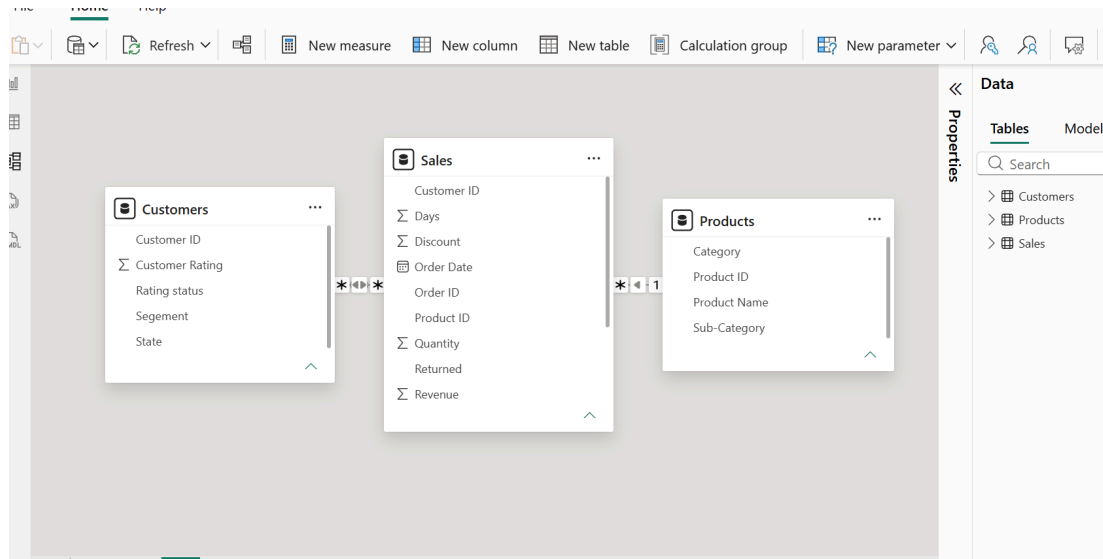
Comments

Add-ins

M2

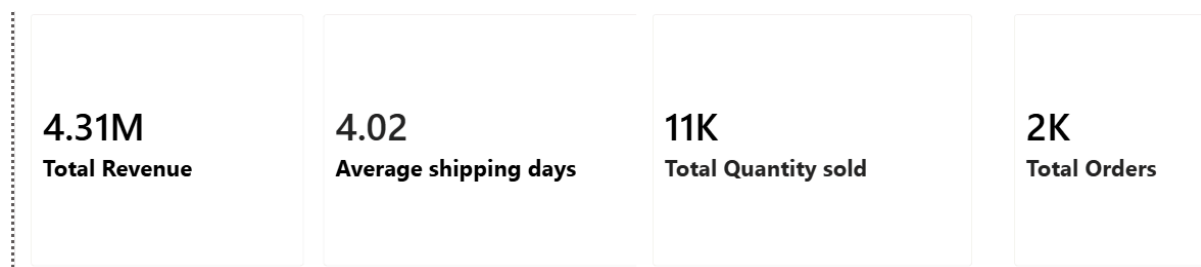
<

**8 Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.



First, I created data relationships between the Customer, Sales, and Product tables in Power BI and checked that the relationships were active

I have created new Cards using measures to display Total Revenue, Average Shipping Days, Total orders and Total Quantity Sold



Created a pie chart to show the revenue distribution across different states

Created a clustered column chart to analyze revenue across different categories and sub-categories

Created a donut chart to show the revenue distribution across different customer segments.

Added a tooltip to display Revenue and Discount details when hovering over the chart

Created a line chart to compare monthly revenue for returned and non-returned orders

Created a clustered bar chart to compare revenue across different ship modes based on return status

Created a stacked column chart to compare quantity sold across categories and sub-categories



## 9. Insights & Conclusions

### Key Findings:

- Identified monthly sales trends.
- Found the best and lowest performing categories.
- Compared sales and profit across different categories.
- Analyzed orders, returns, and shipping modes.

**Descriptive Analysis:**

Analyzed sales, profit, quantity, and orders.

**Diagnostic Analysis:**

Identified the reasons for differences in sales and profit.

**Predictive Analysis:**

Used past sales trends to understand future sales possibilities.

**Prescriptive Analysis:**

Suggested focusing on high-performing products and reducing returns.

**10. Conclusions**

The analysis helped to understand sales, profit, orders, and category performance. Power BI made it easy to visualize the data and identify useful business insights.